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Gamification Frameworks for Behavioural Change Using Psychophysiological Feedback

Ai Salt
Sir Peter Rigby Digital Futures Institute
Aston University
Birmingham, UK
asalt23@aston.ac.uk

Seif Allah El Mesloul Nasri
Sir Peter Rigby Digital Futures Institute
Aston University
Birmingham, UK
nasris@aston.ac.uk

Manolya Kavakli-Thorne
Sir Peter Rigby Digital Futures Institute
Aston University
Birmingham, UK
m.kavakli-thorne@aston.ac.uk

***Abstract*—This paper presents a psychologically grounded framework for gamified learning platforms designed to enhance user engagement and promote prosocial behaviours, particularly in the context of environmental sustainability. It addresses a critical gap in current gamification research: the lack of empirically driven models that link specific game design features to measurable behavioural and emotional outcomes. Drawing on established behaviour change theories, the framework supports the creation of learning environments that are emotionally engaging, socially connected, and intrinsically motivating. Key features include AI-powered non-player characters (NPCs), adaptive feedback mechanisms, and collaborative game elements that together encourage sustained, meaningful interaction. A mixed-methods evaluation is planned for future studies, combining self-report surveys with real-time psychophysiological data—such as heart rate (HR), galvanic skin response (GSR), and heart rate variability. A 2D top-down prototype has been developed to operationalise these concepts, featuring an enclosed play area where users trigger responses by interacting with objects as part of task-based challenges. These interaction points are designed to be dynamically adapted for experimental testing of gamification features. The framework aims to provide a scalable model to inform the design of adaptive learning systems that support both meaningful behavioural change and educational outcomes.**

Keywords*—game-based learning, gamification, NPCs, psychophysiological feedback, behaviour change

I. Introduction

Video games and gamification are an important part of the UK's social, economic, cultural and creative settings [1]. According to the Ofcom 2022 Online Report, 60% of UK adults over 15, and 91% of young people between the ages of 3-15 years old play video games [2]. Many of these people were exposed to digital technology from a young age and have an interest in the field of learning-based games and social applications [3] with the aim to find learning-focused engagement that provides them with a source of motivation and engagement [4]. Gamification uses the features inherent to games in situations involving a lack of motivation and learning [5], [6]. Gamifying is not as simple as just adding gaming components to existing applications; it also requires a good design framework and suitable context and narrative [7]. Gamification is a growing tool and trend within education [6], [8] and supporters of this tool state that it can increase engagement, satisfaction, effectiveness and efficiency of students if implemented correctly [9].

Interactive features like online chats and virtual gift exchanges have been found to fulfil users' desire for social connectedness in digital gamified learning experiences [10]. The quality of interactions has been found to be more effective in building rapport with users than the frequency of interactions [11]. Research on social connectedness in game environments has often focused on dynamics between players, via team collaboration, avatars, or story-driven engagement [12]. However, the existing literature indicates that although the presence of a simple NPC can affect the perceptions and connectedness a user feels in a virtual environment, The connections users make with NPCs is not fully understood [13]. Therefore, investigating its potential to satisfy user's social connectedness involves further exploration.

This paper focuses on the use of NPCs as social agents within gamified systems, hypothesising that NPCs play a crucial role in enhancing social connectedness by providing feedback and engagement in game-based learning. There is growing interest in the intersection of gamification and sustainability from both researchers and practitioners. By making sustainability-related challenges more engaging and rewarding, gamification can effectively motivate individuals and organizations to adopt environmentally and socially responsible practices [18].

The remainder of this paper is structured as follows: Section 2 outlines the background and goals of this study, detailing the research questions, hypotheses and theoretical foundations. Section 3 explores framework and methodology of mapping gamification elements to behaviour change and gameplay. This is followed by discussions on challenges, limitations, and future trends.

II. Background and Goals

A. Theoretical Foundations

1) Behaviroual Change

This paper draws on a multidisciplinary foundation of behavioural science to design gamified learning interventions that effectively promote sustainable actions. Six key theories inform the conceptual framework: Maslow's Hierarchy of Needs, Baumeister's Need to Belong, Skinner's Operant

Conditioning, the Theory of Reasoned Action (TRA), Social Cognitive Theory (SCT), and the COM-B Model. Together, they provide a robust basis for understanding motivation, decision-making, and behavioural reinforcement within gamified environments.

The TRA [42] and its extension, the Theory of Planned Behaviour (TPB), highlight the influence of attitudes and social norms on intentions and behaviours, though do not account for behaviours which aren't controlled by decisions. Similarly, SCT [43] emphasises learning through observation and interactions between personal, environmental, and behavioural factors, making it helpful for understanding external stimuli in gamified systems. However, it can lack specificity in addressing individual engagement and internal motivations.

Other models like the Trans Theoretical Model (TTM) [44] and Information-motivation-behavioural skills (IMBS) [45] provide structured frameworks for behaviour change. TTM's approach is useful for designing interventions but can oversimplify behavioural progression, especially in gamified contexts. IMBS, is valuable for educational gamification but could overlook environmental and social influences. In contrast, the Behaviour Change Wheel and COM-B Model [46] offer a comprehensive and adaptable framework, integrating capability, opportunity, and motivation as core conditions for behaviour change. However, the complexity of COM-B can make practical implementation challenging. A hybrid approach that integrates the strengths of multiple models may be most effective for designing gamification strategies aimed at behavioural and emotional outcomes.

Figure 1 illustrates the combination of these behaviour change frameworks, including the environmental and personal factors present in TRA, SCT, TTM and IMBS while adding a simplified element of standards and policies that are often prominent particularly in business settings. Majority of the factors from each model can broadly be categories into environmental and personal factors as below.

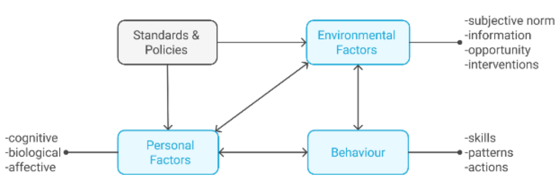

Fig. 1. Combination of behaviour change frameworks

2) *Technology Acceptance Model (TAM)*

The Technology Acceptance Model (TAM), developed by Davis [16], explains how users come to accept and use technology. It identifies perceived usefulness (belief that the system improves performance) and perceived ease of use (belief that the system is effortless to use) as the main predictors of behavioural intention. Rooted in the Theory of Reasoned Action, TAM has been widely applied to digital learning environments [17].

In gamified learning platforms, perceived usefulness relates to how effectively game elements support educational outcomes, while ease of use involves intuitive design and smooth interaction. These perceptions influence learners' willingness to adopt and engage with the platform.

Extended versions of TAM (e.g., TAM2, UTAUT) incorporate factors like social influence, facilitating conditions, and intrinsic motivation—critical in educational settings. This research integrates TAM with psychophysiological feedback to assess real-time engagement, offering a quantitative analysis of how gamification supports technology acceptance and sustained behavioural change.

3) *Evaluation of Existing Gamification Models*

The effectiveness of gamification depends on how well game elements are utilised to meet the outcomes and requirements of the intended context. A gamification framework is a set of guidelines or design principles to provide direction and structure to this process. There are many gamification guidelines, some created specifically for different purposes and industries. Most of these are based on four main frameworks [5] – Octalysis, Mechanics, Dynamics and Aesthetics (MDA), Player-Centred Design (PCD), and 6D.

The Octalysis framework [5] focuses on human motivation, emphasizing emotional engagement and intrinsic drivers like purpose, social influence, and achievement. It offers an adaptable approach suitable across various industries and contexts, but it's strong emphasis on intrinsic motivation may potentially overlook important extrinsic factors that are important in many gamification settings. The complexity of implementing a balance of all eight drives effectively may be challenging and could be considered more conceptual than practical.

The MDA framework [5] provides a clear, structured way to connect game mechanics, player dynamics, and emotional aesthetics, offering clarity on how design choices influence player engagement and motivation. It is also widely used in video game development, making it a well-tested and reliable framework. However, its primary focus is on traditional game design, which may limit its relevance in non-entertainment gamification contexts. The framework also lacks an emphasis on user motivations, psychological elements, and contextual adaptation, which are essential in the purposes of this project.

The 6D Framework [5] offers clear, step-by-step process for designing gamified systems, making it accessible for non-expert implementation. It integrates psychological concepts like player archetypes and types of fun, which helps to adapt designs to various player motivations. It also emphasises practical implementation by aligning goals with user experience. However, reliance on these pre-defined player archetypes may lead to oversimplification of users and overlook more nuanced behavioural motivations. There may also be a risk of prioritising "fun" over desired behavioural change and emotional response outcomes.

The player-centred design framework [5] prioritises a deeper understanding of the user, ensuring that gamification aligns with their needs, motivations, and ethical considerations. It promotes sustainable design by integrating ethical and legal aspects, making it desirable for enterprise contexts where usability and organizational goals are key. However, it does not consider much focus on game mechanics and dynamics, which are important for ensuring user interactivity and engagement. The enterprise-specific focus may also restrict its applicability to broader or consumer-oriented gamification efforts, reducing flexibility in further development for applications in industries such as education or entertainment.

Each framework provides unique and important insights and tools for gamification design that are suitable for different needs and contexts. The Octalysis Framework leverages intrinsic motivations, while the MDA Framework focuses on creating engaging emotional experiences. The 6D Framework provides a practical, step-by-step guide, and the Player-Centred Design Framework is particularly effective in enterprise-specific applications. For a robust and well-rounded gamification strategy, I will combine elements of these frameworks while addressing their limitations to maximize effectiveness in this project and consider potential future developments across varied scenarios.

Figure 2 illustrates the combination of these frameworks into a pyramid structure, where the three base blocks represent the driving design decisions behind gamification applications, incorporating key elements from PCD and 6D and MDA frameworks. Moving upwards, choice of game mechanics and gamification elements are applied in MDA, 6D and PCD frameworks.

Elements of the Octalysis framework can be incorporated into the design using gamification elements, for example through fear, rewards, and storytelling.

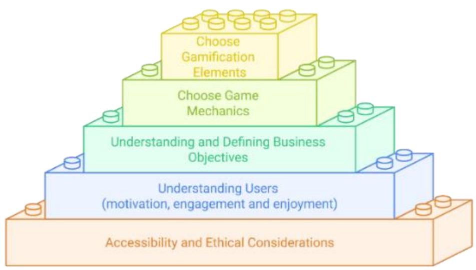

Fig. 2. Gamification strategy pyramid: a combination of gamification frameworks

B. *Research Questions and Hypotheses*

Research Question 1: What are the specific effects of non-player characters (NPCs), social connectedness, and feedback through rewards on users' engagement, intended behaviour, and emotional response in gamified learning platforms?

Hypotheses:

- H1a: AI-driven NPCs will significantly increase user engagement (measured by session duration, retention rate) compared to static instructional content.
- H1b: NPC and feedback features will positively correlate with emotional response measures (EDA, HR, HRV).
- H1c: Structured reward systems will increase completion of pro-environmental tasks (measured by conversion rate).

Research Question 2:

How do we evaluate the effectiveness of a particular gamification feature on users' emotional response, engagement, and intended behavioural change?

Hypotheses:

- H2a: Multimodal assessment combining self-report and physiological measures will provide more accurate prediction of behavioural intentions than either method alone.
- H2b: Real-time physiological feedback will show significant correlation with self-reported engagement measures.
- H2c: Changes in physiological measures during specific gamification interactions will predict subsequent behavioural intentions.

III. Framework and Methodology

A. *Mapping Gamification Elements to Behaviour Change*

This paper adopts a framework that focuses on mapping AI-driven non-playable characters (NPCs) to measurable behavioural and emotional outcomes.

1) *AI-Driven Non-Playable Characters (NPCs)*

In gamified learning environments, NPCs can serve as social stand-ins, enhancing social connectedness and supporting task completion [38, 39]. For example, an NPC representing an endangered animal might deliver real-time feedback on a user's sustainable choices, thereby reinforcing prosocial behaviour. This paper investigates how NPC interactions influence emotional engagement and behavioural intent. Experiments will compare scenarios with no NPC, static NPCs, and adaptive NPCs across multiple participant groups. Physiological data (e.g., heart rate variability, galvanic skin response) will be analysed alongside post-task surveys to understand how NPC interactions affect empathy, trust, and motivation.

Feedback mechanisms are central to gamification's capacity to reinforce behaviour. These can be immediate or delayed, visual or verbal, and quantitative or qualitative [40]. In this experiment, feedback will be embedded through real-time progress indicators, visual reactions from NPCs, and narrative consequences based on user decisions. This is guided by Skinner's operant conditioning [41], where reinforcement (e.g., points, badges) increases the likelihood of repeated desired behaviours.

The methodology integrates user-centred design, physiological feedback, and behavioural theory to map gamification elements to measurable outcomes. This will inform a scalable and evidence-based framework for designing impactful gamified systems aimed at sustainability-focused behavioural change.

2) *Integration of Gamification Elements with Behavioural Theory*

Effective gamification design benefits significantly when its components are explicitly aligned with established behavioural theories. For instance, Self-Determination Theory highlights that intrinsic motivation is enhanced when three core psychological needs—autonomy, competence, and relatedness—are satisfied [20]. In practical terms, this translates into providing users with meaningful choices (addressing autonomy), structured feedback for skill development (addressing competence), and incorporating collaborative or narrative elements (addressing relatedness). Social Cognitive Theory further supports this by emphasising observational learning and self-efficacy [21]. Adaptive non-player characters (NPCs), functioning as mentors or role models, can effectively demonstrate desirable sustainable behaviours, thereby increasing a learner's confidence and motivation to emulate such actions. Maslow's Hierarchy of Needs also provides valuable insights, suggesting gamified environments should cater to fundamental psychological needs: social elements like leaderboards and collaborative tasks fulfil the need for belonging, whereas achievement mechanics such as badges and level advancements satisfy esteem needs [22]. The COM-B model similarly provides a structured framework: challenging tasks and tutorials enhance Capability through skills and knowledge; social interactions and contextual features offer Opportunity; and reward mechanisms sustain Motivation [23]. These theoretical connections clarify the efficacy of specific gamification strategies. Points, badges, and leaderboards serve as reinforcements through operant conditioning [24]. Quantitative point systems, in particular, have been noted to actively encourage participation by rewarding targeted behaviours [25]. Furthermore, narrative and thematic elements foster purposeful contextual learning, deepening user engagement and facilitating memory retention [26]. By intentionally associating each gamification element with these psychological drivers, designers can craft environments that not only engage users but also foster enduring behavioural changes [20, 21, 23].

3) Adaptive Physiological Feedback Systems

Contemporary gamified systems increasingly employ real-time psychophysiological data to dynamically adjust game feedback and difficulty levels, effectively establishing a biofeedback loop. In such closed-loop designs, continuous monitoring of physiological signals, such as heart rate and galvanic skin response (GSR), provides immediate input to the game engine. This system then adapts NPC behaviour, challenge intensity, or narrative progression accordingly. For example, if a player's GSR indicates heightened stress or frustration, the system may simplify subsequent tasks or prompt an NPC to offer supportive feedback. Conversely, indications of reduced arousal or engagement can trigger increased task difficulty or stimulating interactions. This approach, known as biocybernetic adaptation, has been effectively demonstrated in previous studies; for instance, Ewing et al. utilised real-time EEG feedback to optimise player engagement by adjusting difficulty levels in accordance with neural indicators of attention [27]. Recent reviews further highlight the potential of adaptive systems sensitive to emotional states such as anxiety or stress, advocating for more nuanced emotional adaptation [28, 29].

Mercer and De Franches underscore that biofeedback-integrated games can significantly enhance emotional awareness and cognitive control [29]. Within the proposed framework, NPC interactions are dynamically adapted based on real-time monitoring of physiological markers, such HRV and GSR. This dynamic adaptation ensures users remain in an optimal state of engagement, fostering sustained motivation and improved learning outcomes [27, 28, 29, 30].

Figure 2 presents the closed-loop architecture that underpins the adaptive gamified system proposed in this work. In this setup, the learner's physiological data are captured in real time via wearable sensors and transmitted to the game engine. This engine processes the data using a dedicated parser and an adaptation algorithm, which together determine how the system should respond. These outputs are then used to guide the actions of an AI-driven NPC, which can adjust its feedback dynamically.

Depending on the learner's physiological state, the NPC may offer motivational support, verbal cues, or modify the challenge level of the task. This adaptive feedback is intended to help maintain the learner's engagement, optimise cognitive effort, and support affective regulation. The system completes the loop by continually monitoring the learner's state and updating its responses accordingly, ensuring the interaction remains responsive and personalised throughout the experience.

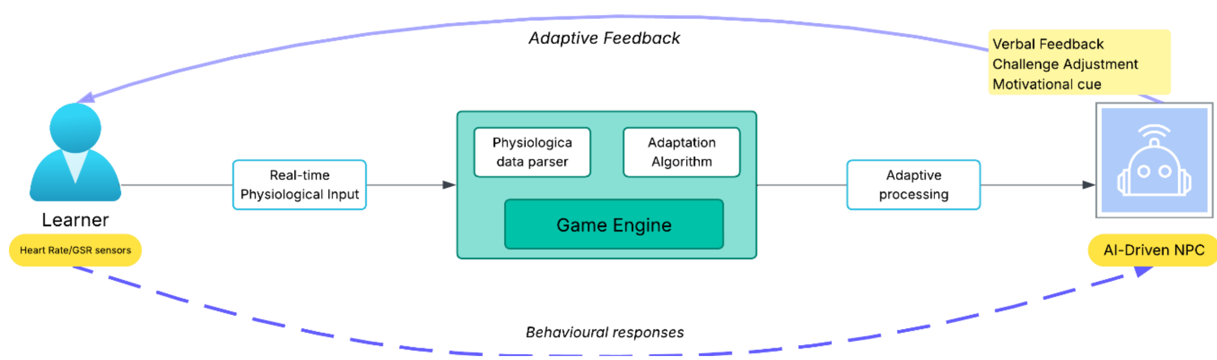

Fig. 3. Closed-Loop NPC Feedback System.

4) Game Design

The proposed game combines an informational task-based puzzle with stealth gameplay, designed as a platform to explore the efficacy of gamification in promoting awareness of environmental stewardship. The game adopts a structured methodology that integrates NPC-driven interactions, dynamic feedback mechanisms, and interactive learning pathways to engage players in solving mini-puzzles and completing a to-do list centred on environmental standards.

Quizzes embedded within the gameplay challenge players to critically evaluate how businesses can adopt innovative sustainability practices, address global climate and social challenges, assess carbon reduction strategies, and analyse the objectives of international programs for long-term sustainability. Upon completing the game, players receive a personalised impact report summarising their decisions, providing actionable recommendations for reducing energy costs and accelerating organisational green strategies.

Figure 4 Game Design Flow illustrates the structural flow of the game, demonstrating its progression from the introduction of objectives to NPC-guided tasks, interactive learning activities, and real-time gamification feedback, culminating in the delivery of a personalised sustainability report. The game is anticipated to contribute to increased comprehension of environmental standards, improved critical thinking regarding sustainability challenges, and enhanced practical decision-making that supports organisational decarbonisation efforts.

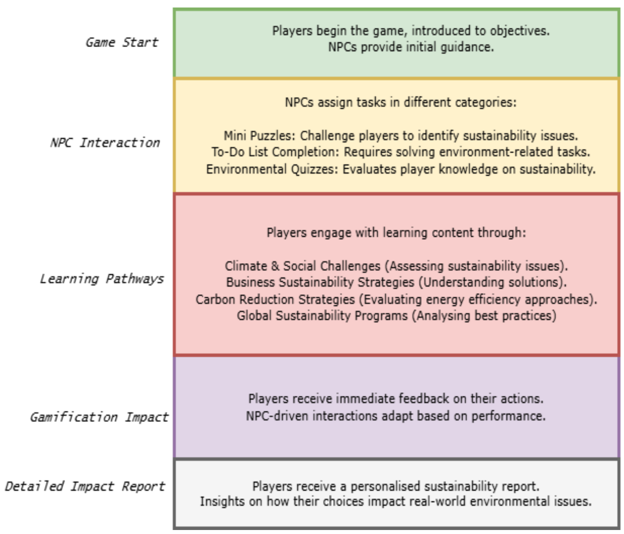

Fig. 4. Game Design Flow

The art and visual style will be basic 2D cartoon/stylized art with block colouring. Figure 5 Mood board depicting art style, look and feel is a mood board illustrating the desired look and feel of the game that will be developed, with some basic annotations describing these styles.

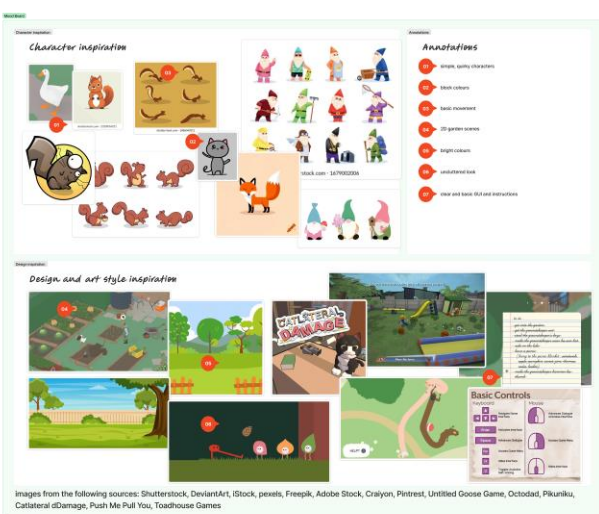

Fig. 5. Mood board depicting art style, look and feel

5) Experimental Procedure

Based on initial survey data and participant demographics, the system will be tailored either to a general public audience or employees within small and medium-sized enterprises (SMEs). A total of approximately 300–360 participants will be recruited through word-of-mouth, SME networks, and sustainability-oriented organizations. Participants will complete a pre-experiment survey assessing baseline attitudes, knowledge, motivation regarding sustainability, and gaming experience. They will then be randomly assigned to experimental groups.

The experiment includes three treatment conditions. Data will be collected using web analytics, self-report instruments, and psychophysiological measures (e.g., heart rate, galvanic skin response) [19].

The experiment will compare no NPCs, passive NPCs, and interactive NPCs to assess engagement and task performance.

Post-experiment data collection includes follow-up surveys to assess changes in sustainability awareness and motivation, along with usability and relevance ratings of game elements. A subset of participants will also be invited to take part in interviews or focus groups for qualitative insights.

B. Mapping Gamification Elements to Gameplay

As outlined in the experimental procedure above, we plan to conduct several experiments to map gamification elements to gameplay components.

1) Characters (NPCs)

NPCs (AI-driven or scripted) can serve as guides, peers, or challengers, facilitating feedback and fostering social connectedness. This experiment will use adaptive NPCs capable of emotional feedback to simulate interpersonal interaction. We will conduct A/B testing of presence and absence of NPCs to assess impact on motivation and emotional response using post-play surveys and physiological engagement metrics (e.g., GSR, HRV).

2) Environments (Virtual Worlds, Contexts, Themes)

Environments can provide narrative context and sensory cues that influence users. Themed settings (e.g., a polluted world vs. a sustainable utopia) reinforce values and consequences. Contextual cues (like visual degradation or growth based on user behaviour) serve as feedback, and adaptive environments respond to user decisions, reinforcing agency and responsibility. We will test parallel versions of the platform with different environmental contexts to incorporate real-time environment changes based on user choices (e.g., cleaner air as users complete sustainability tasks), and track engagement via time-in-environment metrics, physiological feedback, and self-reported post play surveys.

3) Interactions, Missions, and Events

Interactions, missions and events can define gameplay progression, decision-making, and pacing. They are essential for delivering feedback, establishing goals, and maintaining flow. Quests/missions act as structured learning tasks with immediate feedback loops (operant conditioning). Collaborative tasks and competitive events increase social presence and motivation, and dynamic events (e.g., community clean-up missions) can simulate real-world impact and urgency. To test this, we will separate missions into types (informational, cooperative, competitive) and test their effects on motivation and performance by logging user decision paths and use surveys and psychophysiological data (e.g., skin conductance) to evaluate engagement peaks during key events.

Each gameplay component (Character, Environment, Interaction) contributes to emotional response, learning and behavioural reinforcement. They can be mapped to target outcomes such as behavioural change, engagement, and sustainability awareness, and measured using a mix of pre/post surveys, game metrics, biofeedback (e.g., HR, GSR) and interviews.

Figure 6 incorporates each hypothesis into a TAM model, illustrating the connections described above. The gamification elements are grouped together as external variables, and perceived usefulness and perceived ease of use are grouped as self-reported metrics that will be collected as part of the participant surveys. Quantitative data metrics are grouped together, measuring engagement and emotional response via game metrics and psychophysiological feedback. Each hypothesis as outlined in Section 2 is represented as arrows depicting the relationship between either individual elements or grouped elements in the model.

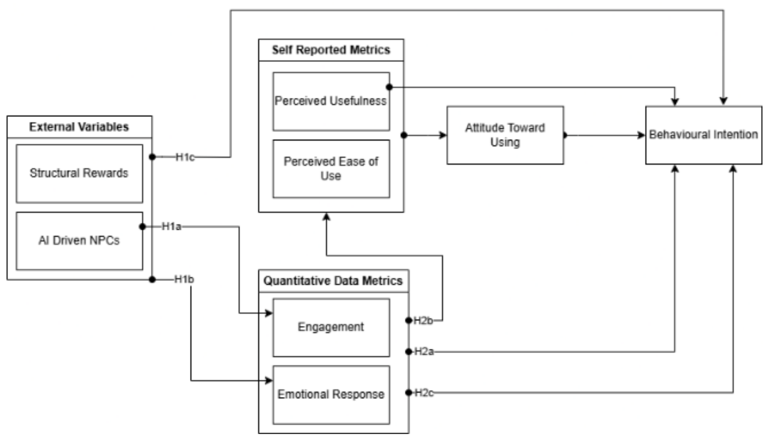

Fig. 6. Research model for technology acceptance

4) Learning and Cognitive Outcomes of Gamified Interventions

In addition to motivational and behavioural enhancements, gamified learning interventions can yield substantial cognitive and educational improvements.

Meta-analyses demonstrate that gamification significantly enhances knowledge retention, problem-solving capabilities, and academic performance. For example, Sailer and Homner found a moderate positive impact on cognitive learning outcomes, with an overall effect size (g) of approximately 0.49 when comparing gamified interventions to traditional instruction [31]. These improvements are largely attributed to active learning processes inherent in gamified tasks, such as quests and puzzles, which encourage the practical application of learned concepts, thereby reinforcing comprehension. Integrating narrative elements and realistic scenarios helps embed new information within familiar cognitive schemas, facilitating enhanced memory retention [26, 32]. Immediate feedback provided within game contexts, such as point systems and NPC reactions, allows rapid correction of misconceptions, supporting deeper cognitive processing consistent with Cognitive Load Theory principles [33]. Additionally, well-structured gamified tasks have been shown to effectively reduce extraneous cognitive load by introducing complexity progressively, thus preventing the overload of working memory [34]. Furthermore, social and collaborative tasks engage multiple cognitive processes, including communication, strategic planning, and executive functions, promoting better skill transfer through contextualised learning practices [35]. Therefore, engaging game mechanics coupled with detailed feedback are anticipated to enhance cognitive engagement, including attention, persistence, and reflective thinking, alongside tangible learning outcomes, such as knowledge acquisition and practical application [31, 32, 35].

IV. Prototype

Figure 7 shows an initial prototype of a 2D top-down style gameplay. There is an enclosed play area that users will navigate while completing different tasks. The white symbols on the image decpict every trigger point in the game where a user interaction is performed or responded to.

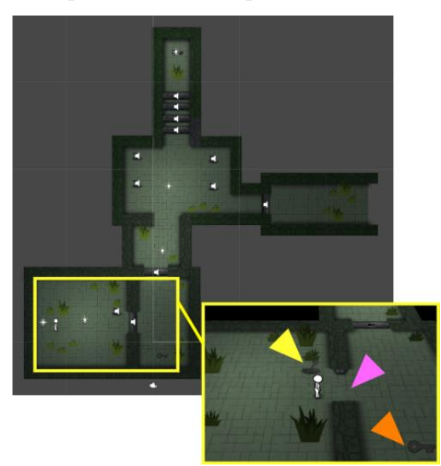

Fig. 7. Prototype

The yellow arrow points to an example of an action trigger – a lever for the user to pull. The pink arrow points towards a gate or barrier that responds to the user action. The orange arrow points towards a different kind of user interaction, where the user will pick up an object to carry it to another trigger point as a task. This prototype outlines the basic game mechanics and interactions points. Each trigger and trigger response will be adapted to experimental needs to test specific game elements accordingly.

V. Ethical Considerations

Conducting research on gamification models and frameworks using psychophysiological feedback requires careful consideration of ethical implications to ensure participant well-being, data integrity, and responsible practices. The use of in-game rewards must prioritise ethical design to avoid exploiting psychological vulnerabilities, such as variable reward schedules that could potentially lead to compulsive behaviours, gambling or addiction [47]. As a researcher I will adopt responsible data collection and sharing practices, adhering to data privacy, anonymity, and security standards outlined in frameworks such as the UK government video games research framework [1]. This includes obtaining informed consent, implementing data protection measures, and safeguarding against misuse of personal data. All participants should be fully informed about the nature of the research, the type of data being collected, and the intended use of their data. This will ensure that participants are aware of any potential risks or benefits associated with the study. Accessibility and inclusivity should also be emphasised to accommodate diverse participant needs. This will involve designing interfaces that are user-friendly and accommodating to participants with disabilities, thereby promoting inclusivity in research.

VI. Challenges and Limitations

Developing and testing an effective gamification framework presents several challenges and limitations. Designing models that effectively combine engagement with learning requires a balance between being interesting and not appearing overly patronizing. Tailoring content to diverse target audiences is also demanding, as different groups may respond differently to gamified elements and have different baseline knowledge of the target material. Psychophysiological feedback, while insightful, requires specialised equipment and expertise for data collection, which can complicate testing. Limited sample sizes, due to difficulties in recruiting volunteers, can further negatively affect the reliability of results, and online releases risk users tampering with the system, compromising data integrity.

VII. Future Trends and Research Directions

Future research on gamification models and frameworks using psychophysiological feedback offers exciting possibilities across various domains. Beyond sustainability and more commonly known environmental sectors, gamification could expand into game-based learning for other environmental sectors such as rivers, oceans, marine ecosystems, and waste management.

Exploring other psychophysiological feedback, including EEG and brain-controlled interfaces could also open exciting avenues into gamification research. Incorporating AI and machine learning could vastly improve potential to personalise gamified experiences, adapting content dynamically to individual preferences and performance and allowing further exploration into the specific effects these design decisions have on user emotional and behavioural response. Additionally, the framework could be translated to advanced technologies like mixed reality (MR), virtual reality (VR), augmented reality (AR), and immersive environments such as CAVEs or motion-enabled platforms, enhancing engagement through more holistic sensory immersion. Integration with business systems, including assistive technologies, content management systems, and enterprise resource planning software, could create unified solutions that align gamification with broader organisational objectives. These advancements could enhance gamification's effectiveness and expand its application, driving innovation and growth in industries beyond its current scope

VIII. Conclusion

This paper has presented a novel framework for gamified learning platforms that integrates psychological theory, technology acceptance modelling, and physiological measurement to promote engagement and pro-environmental behaviour. Through the investigation of AI-driven NPCs, social connectedness features, and structured reward systems, we address critical gaps in gamification literature—particularly the lack of data-driven models demonstrating measurable behavioural outcomes.

Our key contributions include: (1) an extended Technology Acceptance Model that maps gamification elements to specific physiological and behavioural outcomes; (2) a methodologically robust approach to evaluating engagement through real-time physiological feedback; and (3) a framework demonstrating systematic application of behavioural change theories to gamification design.

The broader impact extends beyond environmental education. Our findings on AI-driven NPCs and closed-loop adaptive architecture address areas for advancement in online learning environments and technology. Our approach to mapping gamification elements to psychological constructs offers a template for more rigorous evaluation, shifting gamification research from descriptive case studies to evidence-based design science.

This research demonstrates that gamification, when thoughtfully designed with solid theoretical foundations and rigorously evaluated through multiple data streams, can serve as a powerful tool for enhancing learning and fostering prosocial behaviours needed to address real-world sustainability challenges.

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